

Proteolysis Consistent with Activation of Caspase-7 after Severe Traumatic Brain Injury in Humans

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ABSTRACT

The expression and proteolysis of caspase family proteins are involved in the initiation and execution of apoptosis, which has been reported to occur in human and experimental traumatic brain injury (TBI). Caspase-3, -6, and -7 belong to the group of executioner caspases, which are cleaved and activated at the late, irreversible stage of apoptosis. Our previous studies demonstrated roles for caspase-1, -3, and -8 in humans after severe TBI. Here we report expression of caspase-7 mRNA and protein in humans after TBI ($n = 16$) and control brain-bank tissue ($n = 6$). Semiquantitative reverse transcription polymerase chain reaction showed no differences between caspase-7 mRNA in TBI patients versus controls (73 ± 24 vs. 85 ± 56 relative optical density [ROD], respectively). In contrast, Western blot analysis showed increased pro-caspase-7 in TBI patients versus controls (214 ± 30 vs. 1 ± 1 ROD, respectively), as well as an increase in the ~ 20 kD proteolytic fragment in TBI patients versus controls (86 ± 13 vs. 22 ± 12 ROD, respectively), consistent with activation of caspase-7 after TBI in humans. Immunohistochemical analysis showed that cells expressing caspase-7 included astrocytes and neurons and possibly other glial cell types and infiltrated inflammatory cells. These data show that caspase-7 and its cleavage product are increased in human brain after TBI in many central nervous system, as well as noncentral nervous system, cell types. Thus, caspase-7 may play a role in the glial and inflammatory responses, and possibly neuronal death, after TBI in humans.

Key words: apoptosis; cysteine protease; head injury; inflammation; programmed cell death

INTRODUCTION

NEURONAL DEATH after traumatic brain injury (TBI) involves a component of programmed cell death or apoptosis, an evolutionarily conserved mechanism for whole cell clearance that limits local and systemic in-

flammatory reactions. The activation of caspases, a family of cysteine (thiol) peptidases, in injured neurons has been identified to be a critical event in programmed cell death in various neurological diseases, including TBI, stroke, and epilepsy (Liou et al., 2003). Initiation of the caspase cascade may occur through oligomerization of

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cell surface death receptors or by mitochondrial signals (Eldadah and Faden, 2000). Both mechanisms converge at the point of activation of "executioner" caspases, which include caspase-3, -6 and -7. Caspase-3 is the most completely characterized with a demonstrated role in experimental models of TBI (Yakovlev et al., 1997; Clark et al., 2000), cerebral ischemia (Namura et al., 1998), and epilepsy (Henshall et al., 2000a). A role for caspase-3 is also suggested in human TBI (Clark et al., 1999) and epilepsy (Henshall et al., 2000b). Caspase-6 is important in the proteolysis of nuclear lamin, likely related to morphologic changes in the nucleus during apoptosis (Ruchaud et al., 2002).

The human *caspase-7* (MCH-3, ICE-LAP3, CMH-1, SCA-2, LICE2) gene is located on chromosome 10q25 and yields 11 potential transcripts by alternative splicing predicted to encode 11 distinct proteins (Duan et al., 1996). The sizes of these full-length translated proteins range from 34.1 to 42.7 kDa. The full-length inactivated zymogen can be cleaved by caspase-3, -6, or -10 to p20 and p12 subunits, permitting activation of the enzyme. Caspase-7 is often categorized with caspase-3 because they share the same cleavage site, thus providing redundancy to caspase-3 in the execution of programmed cell death. However, a dominant role for caspase-7 in non-developmental neuronal apoptosis remains somewhat controversial. A recent publication has demonstrated increased caspase-7 expression and proteolysis after TBI in rats (Larner et al., 2005). In addition, several caspase-7-specific substrates have been identified, including those important in protein synthesis and degradation, supporting unique functions for active caspase-7 in disease. In the present study, we examined caspase-7 mRNA and protein expression in brain tissue samples from humans after severe TBI.

MATERIALS AND METHODS

Human Brain Tissue Samples

The study protocol was approved by the Institutional Review Board at the University of Pittsburgh. Informed consent was obtained from the legal next of kin for each patient; approval was granted for existing tissue specimens. Patients in the study were admitted to the University of Pittsburgh Medical Center between August 1995 and January 2000. Brain tissue samples were obtained from patients who underwent a decompressive craniectomy as part of the management of life-threatening intracranial hypertension after TBI (defined as clinical or radiographic evidence of cerebral herniation or impending cerebral herniation, or significant mass effect). The samples were collected after surgical resection of the in-

jured area of brain; the samples were of residual tissue that would otherwise have been discarded. Samples were primarily from injured temporal lobe cortex and include samples used in previously published studies (Clark et al., 1999; Zhang et al., 2003). Tissue was frozen and stored at -70°C until used for analysis. Control samples were frozen postmortem brain-bank specimens obtained from patients who died of causes not related to CNS trauma. Control samples were from temporal lobe cortex and were originally collected between April and October 1996.

RNA Extraction and Reverse Transcription Polymerase Chain Reaction

Total cellular RNA was extracted from brain tissues using Trizol reagent (Life Technologies, Gaithersburg, MD). Total RNA concentration was determined using a spectrophotometer (260 nm). For patient C, the mRNA in the sample was undetectable and presumably degraded, so it was not used for polymerase chain reaction (PCR). One μg of total RNA from each sample was reverse transcribed with 200 U of Superscript II reverse transcriptase (Invitrogen, Carlsbad, CA) for 50 min at 42°C and primed with 0.5 μg of random hexamer under the conditions recommended by the manufacturer. Reverse transcription (RT) products were then digested with RNaseH at 37°C for 30 min to remove the RNA template from the cDNA:RNA hybrid molecule. One μL of each cDNA sample was amplified by PCR using primers targeting human *caspase-7* (5'-AGTGACAGGTATGGGCGTTC-3' and 5'-CGGCATTTGTATGGTCCTCT-3') and β -actin (proprietary primers purchased from Ambion, Austin, TX). PCR was performed in a Robocycler Gradient 96 PCR system (Stratagene, La Jolla, CA). Reagents were assembled in a final volume of 50 μL , the final concentration of reagents being as follows: 1 μL of first-strand cDNA, 1 μM of each primer, $1\times$ PCR amplification buffer, 2.5 mM MgSO_4 , $1\times$ dNTP mix (Roche Molecular Biochemicals, Indianapolis, IN), and RNase free water to 50 μL . Samples were initially denatured at 94°C for 3 min, followed by the addition of 2 U of platinum Taq DNA polymerase (Life Technologies). PCR conditions were first optimized to determine the number of cycles for reaching the exponential stage phase of amplification. Thermocycling parameters were 1 min at 94°C , 1 min annealing at 55°C , and 1 min extension at 72°C , repeated for 30 cycles. PCR products were then separated on a 1.2% agarose gel. PCR products were visualized by ultraviolet (UV) light after ethidium bromide staining. The gels were digitized, and the relative optical density (ROD) of the bands of interest were semiquantified using a Kodak Image Station (Kodak 440CF, Rochester, NY).

CASPASE-7 ACTIVATION AFTER TBI

Western Blot Analysis

Human brain tissue samples were sonicated in lysis buffer containing 0.1 M NaCl, 0.01 M Tris, 0.1mM EDTA, and the protease inhibitors chymostatin (2 μ g/mL), leupeptin (2 μ g/mL), pepstatin (2 μ g/mL), and phenylmethylsulphonyl fluoride (100 μ g/mL). Lysates

were centrifuged at 10,000g for 30 min at 4°C; then protein concentration was determined using Bicinchoninic acid (Pierce, Rockford, IL). Fifty μ g protein samples were then boiled in loading buffer for 5 min, followed by chilling on ice. Protein samples were separated on a 12% SDS-PAGE gel and transferred onto polyvinyl di-fluoride membranes (Bio-Rad, Hercules, CA). After

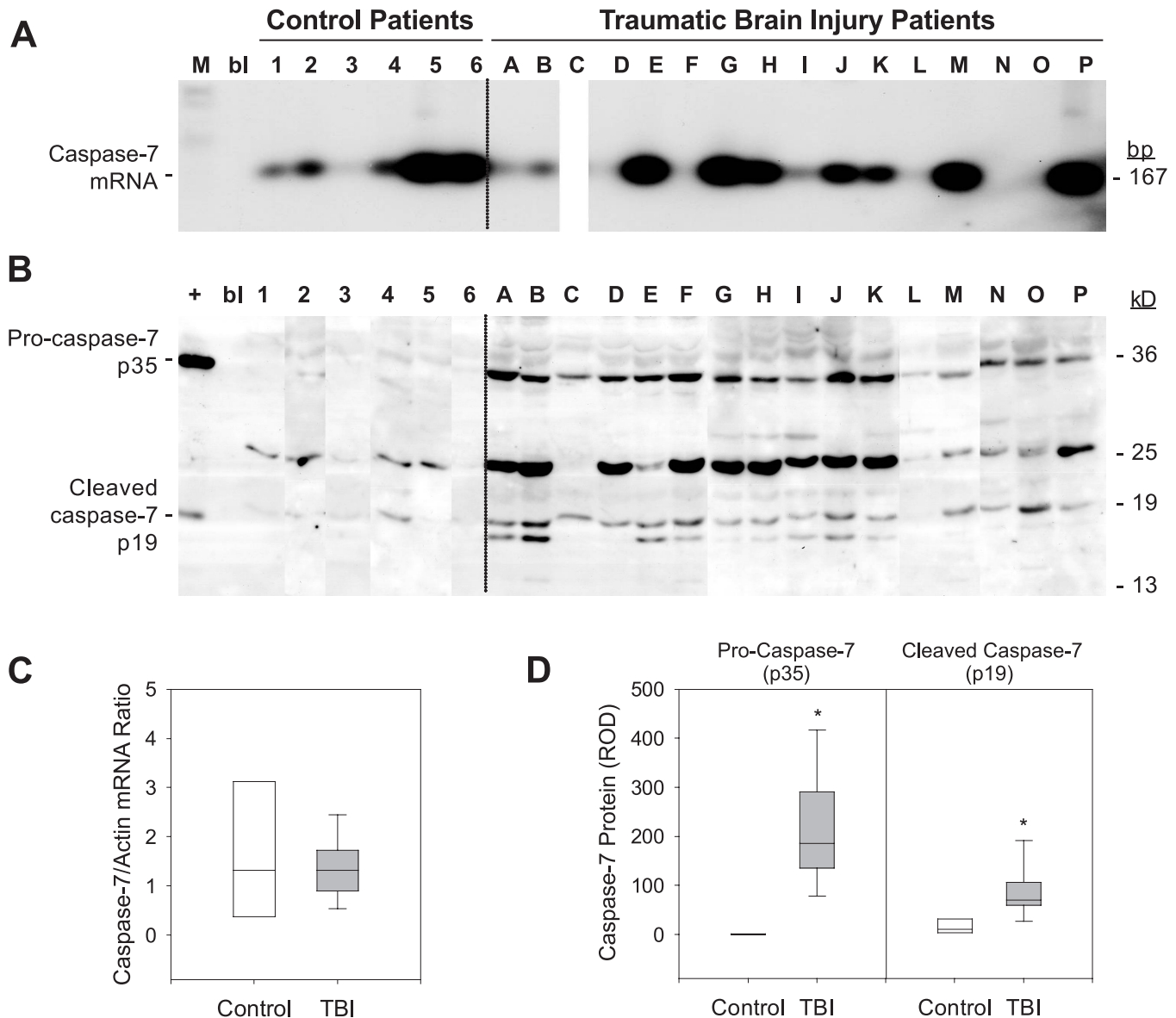


FIG. 1. Expression of *caspase-7* mRNA and protein in human brain samples from patients after TBI and from brain-bank post-mortem control patients. **(A)** RT-PCR showing a predicted 167 bp *caspase-7* PCR product. *Caspase-7* mRNA was detected in the majority of TBI and control patients. **(B)** Western blot analysis showing increased pro-caspase-7 (p35) and cleaved caspase-7 (p20) in TBI versus control patients. **(C)** Caspase-7/ β -actin ratios in TBI and control patients. There was no difference between groups ($p = 0.907$). **(D)** Relative optical density (ROD) of p35 and p20 peptide bands in TBI and control patients. Both p35 and p20 peptide bands were increased in TBI versus control patients ($*p < 0.01$). M, marker; bl, blank; +, positive control.

blocking overnight in 5% nonfat milk in PBS, the membrane was incubated with goat polyclonal antibody against the p20 subunit of human caspase-7 (clone N-17, SC-8510; Santa Cruz Biotechnologies, Santa Cruz, CA), diluted 1:200 in blocking reagent at room temperature for 1 h, and washed three times in 0.1% Tween-20 in PBS for 10 min. Secondary antibody was applied at a 1:2000 dilution for 1 h. The blot was washed in PBST, incubated with a chemiluminescence detection system (NEN Life Science, Boston, MA), and exposed to x-ray film. The RODs of the detected peptides were quantified using a Kodak imaging system.

Immunohistochemistry

To determine whether caspase-7 was localized to neurons, immunohistochemistry was performed using frozen sections from five randomly selected TBI patients. These specimens were immersion-fixed in 2% paraformaldehyde, cryoprotected in sucrose, and then snap frozen in 2-methylbutane in liquid nitrogen. Since control tissue processed in this manner was not available, samples from control patients were not processed for immunohistochemistry. Specimens were cut in 10 μ M sections using a cryostat and mounted on glass slides. Slides were washed three times with PBS containing 0.5% bovine serum albumin and 0.15% glycine (buffer A). Nonspecific activity was blocked with 5% normal goat serum in buffer A. Brain sections were then incubated overnight at room temperature in a 1:100 dilution of caspase-7 antibody. Consecutive sections were incubated in a 1:500 dilution of antibody against the neuron-specific marker NeuN (Chemicon, Temecula, CA). Sections were then washed three times in buffer A and incubated for 1 h in a 1:200 dilution of streptavidin conjugated Alexa 488 (Molecular Probes, Eugene, OR). Sections were then washed three times in buffer A. Bis-benzimide was applied to tissue prior to cover slipping to identify cell nuclei. Images were collected using an Olympus America Inc. (Center Valley, PA) light microscope. In sections

from each specimen, the primary antibody was omitted to assess for nonspecific binding of the secondary antibody.

To determine whether cells expressing caspase-7 included neurons and/or astrocytes, dual label studies were performed using NeuN to label neurons and glial fibrillary acidic protein (GFAP) to label astrocytes, as previously described (Zhang et al., 2003). Sections were incubated with a rabbit anti-caspase-7 antibody (CT, number 3465; ProSci, Poway, CA) together with either mouse anti-GFAP (Sigma, St. Louis, MO) or mouse anti-NeuN antibody (Chemicon). Sections were then incubated with a mixture of goat anti-rabbit cy3 (Jackson Immunochemicals, West Grove, PA) and goat anti-mouse Alexa 488 (Molecular Probes). Confocal stacks were taken using an Olympus FluoView 500 confocal microscope. In sections from each specimen, primary antibodies were omitted to assess for nonspecific binding of the secondary antibodies.

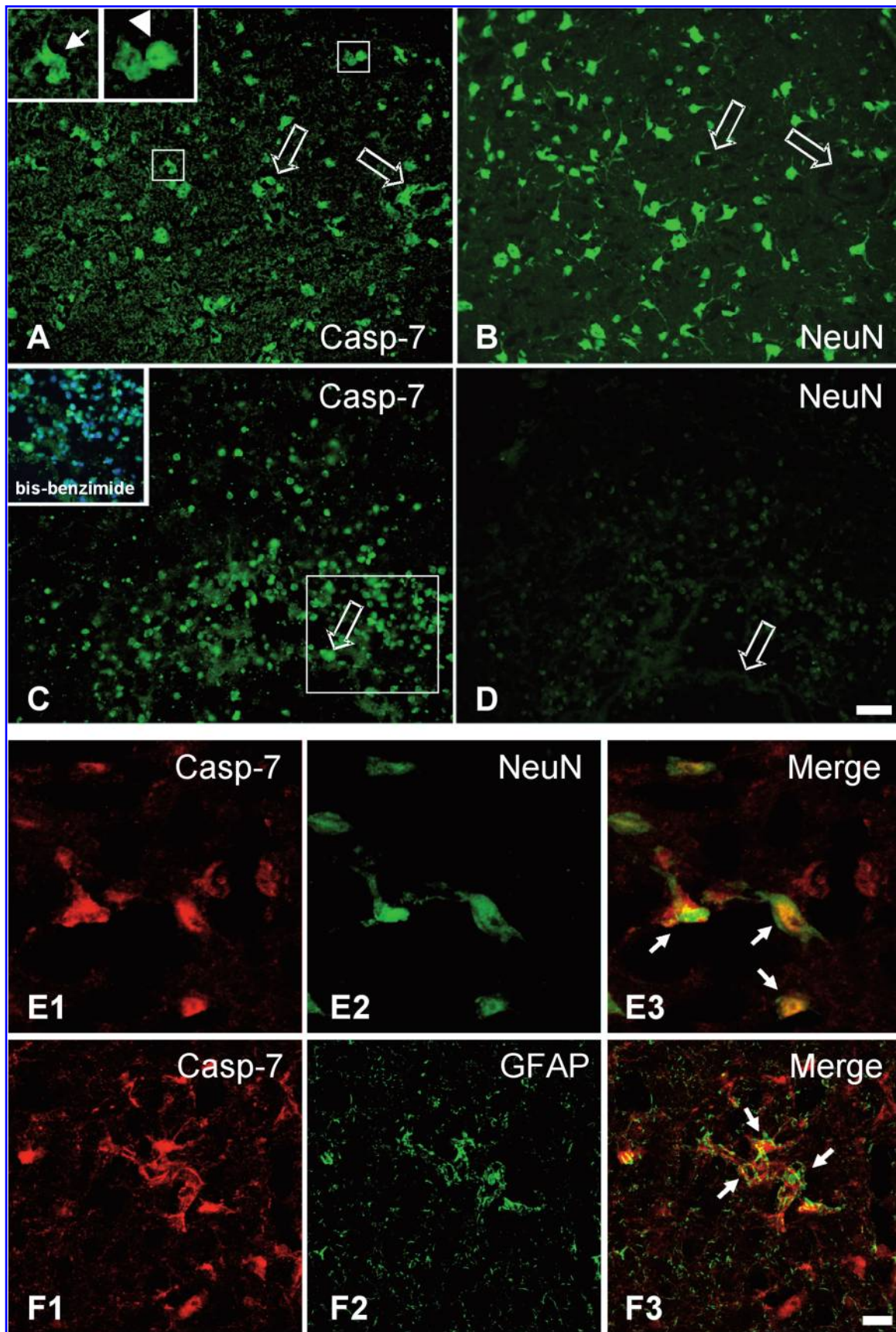
Data Analysis

Data are presented as mean $\pm$ standard error (SE) or median [range] where appropriate. Differences in relative protein and mRNA levels between TBI and control groups were determined using a *t* test or a Mann-Whitney rank sum test if the data violated the assumption of normality and/or equal variance. Potential clinical associations between caspase-7 protein and mRNA and the clinical variables' age, initial Glasgow Coma Scale (GCS) score, gender, and survival were evaluated using Spearman rank correlations or Mann-Whitney rank sum tests where appropriate.

RESULTS

The samples used in this study were from the University of Pittsburgh Brain Trauma Research Center tissue bank. The majority of the samples (6/6 control patients and 16/17 TBI patients) were from patients in whom sep-

FIG. 2. Immunohistochemical localization of caspase-7 in human brain samples from patients after TBI. Two representative patients (of 5 randomly selected) are shown in panels A and B (sample from gray matter) and C and D (sample from white matter). (A and C). Caspase-7 (green) appeared to be localized in infiltrated leukocytes (inset, arrowhead and inset, arrow). Inflammatory loci with many caspase-7 immunoreactive cells were noted in two of five patients (one example shown in C). The inset includes bis-benzimide (blue) labeling of cell nuclei to verify that the immunoreactive cells did not represent nonspecific fluorescence in erythrocytes. Caspase-7 immunoreactive cells were also detected in perivascular locations (open arrows). (B and D). Neuronal labeling with NeuN verified that tissue sections were from gray (A and B) and white (C and D) matter. Dual label studies were used to identify expression of caspase-7 (red, E1 and F1) in neurons using anti-NeuN antibody (green, E2) and astrocytes using anti-GFAP antibody (green, F2). Colocalization of caspase-7 with both NeuN (arrows, E3) and GFAP (arrows, F3) was observed. Note that different fluorescent labels were used to detect caspase-7 in A and C versus E and F. Size bar A–D, 50 μ M; E and F, 10 μ M.



arate tissue was used to report the presence of cell death markers after TBI, including caspase-8 (Zhang et al., 2003). For the control patients, there were four females and two males, age = 52.2 ± 21.3 y, and postmortem time = 10.5 ± 4.3 h. For the TBI patients, there were five females and eleven males, age = 40.4 ± 10.1 , median initial GCS score = 6 [3–15], and time from injury to resection = 17.2 ± 4.4 h. Mechanism of injury included nine motor vehicle-related accidents, five falls, and two assaults. Detailed demographic data can be found in Zhang et al. (2003). There were no clinical associations between *caspase-7* mRNA, pro-caspase-7, or cleaved caspase-7, and the clinical variables' age, gender, survival, or initial GCS (all $p < 0.05$).

Caspase-7 mRNA after Traumatic Brain Injury

As shown in Figure 1A, predicted PCR products of 167 bp for human *caspase-7* were detected. Eight of the 15 TBI patients and five of the six controls showed induction of *caspase-7* mRNA. *Caspase-7* mRNA was similar in brain tissue samples from TBI patients compared with postmortem controls, shown as the *caspase-7*/ β -actin ratio (Fig. 1B; $p = 0.907$). RT-PCR for human β -actin was previously performed in these samples, and their data were previously reported (Zhang et al., 2003). Raw *caspase-7* mRNA values for TBI and control patients were 73 ± 24 and 85 ± 56 RODs, respectively.

Caspase-7 Protein Expression and Cleavage after Traumatic Brain Injury

Western blot analysis of injured human brain samples detected a full-length ~ 35 kD protein band consistent with pro-caspase-7, and an ~ 19 kD protein band consistent with the N-terminal cleaved fragment of caspase-7 (Fig. 1B). Truncated bands ~ 28 kD and ~ 17 kD in size were also detected. Both the precursor forms appear to be higher in TBI patients compared with controls. The relative abundance of both pro-caspase-7 and the p20 caspase-7 fragment was increased in TBI patients compared with control patients (Fig. 1D), with a 163-fold increase in pro-caspase-7 ($p < 0.001$) and a 3-fold increase in the p20 caspase-7 fragment ($p < 0.01$). Cleaved caspase-7 bands were detected in several TBI patients and in one of the controls.

Immunohistochemical analysis showed many caspase-7 immunoreactive cells in injured brain (Fig. 2). The majority of these cells appeared to be astrocytes, other glial cell types, or leukocytes, as consecutive sections alternately labeled with caspase-7 and the neuronal marker NeuN showed an abundance of caspase-7 in both gray matter (Fig. 2A and B) and white matter (Fig. 2C and D).

Dual label studies confirmed the presence of caspase-7 in astrocytes (Fig. 2F). For the most part, colocalization with caspase-7 and NeuN was not apparent when consecutive sections were examined; however, some neurons were found to colocalize with caspase-7 when using the more sensitive technique, dual labeling (Fig. 2E). Caspase-7 immunoreactivity was also associated with brain blood vessels. In two patients, inflammatory foci were detected based on robust caspase-7 immunoreactivity in small, round NeuN-negative cells surrounding blood vessels (Fig. 2C). These cells contained nuclei identified with bis-benzimide (Fig. 2C, inset), demonstrating that these cells were not erythrocytes.

DISCUSSION

A growing body of evidence indicates that mechanisms of neuronal death after TBI in humans involve the activation of caspase family proteases. We previously reported potential roles for caspase-1 and -3 in both pediatric (Satchell et al., 2005) and adult (Clark et al., 1999) patients after severe TBI. More recently, we reported a role for caspase-8 and the Fas-mediated programmed cell death pathway in neurons in adult patients after severe TBI (Zhang et al., 2003). A role for Fas-mediated programmed cell death, related to formation of death-inducing signaling complexes, in human TBI has also been shown by others (Qiu et al., 2002). In this paper, we report a role for the executioner caspase-7 after human TBI; however, in contrast to caspase-3 and -8, caspase-7 expression appears to be primarily nonneuronal and more abundant in glia and infiltrating leukocytes.

The results of the current study indicate that caspase-7 protein is increased and proteolytically processed in brain after TBI, although *caspase-7* mRNA levels were not changed compared with control samples. One possible reason for this discrepancy is transcriptional regulation of *caspase-7*. Consistent with this are the findings that *caspase-7* mRNA is detectable in control, as well as in TBI patients, but caspase-7 protein levels were markedly increased only in TBI patients (Fig. 1). mRNA integrity varied but was intact in the samples, as evidenced by RT-PCR for actin in these patients (see Zhang et al., 2003); however, since the RT-PCR method used was only semiquantitative, these data may not be entirely representative of actual *caspase-7* mRNA levels. It is also possible that the presence or absence of caspase-7 mRNA in some patients is related to the cause of death. An increase in *caspase-7* mRNA has been reported in postmortem brain tissue from elderly patients with severe clinical dementia compared with those who were cognitively normal (Pompl et al., 2003). Full-length *caspase-*

7 mRNA has a long 5'-untranslated region (UTR) rich in guanine and cytosine ribonucleotides, and it is possible that this 5'-UTR may affect translation after TBI. Another possibility is that the source of pro-caspase-7 and cleaved caspase-7 after TBI may be nonparenchymal cells infiltrating into brain (i.e., neutrophils and other mononuclear leukocytes). Consistent with this possibility are the immunohistochemical data, suggesting that a prevalent source of caspase-7 immunoreactive cells could represent leukocytes infiltrating injured brain tissue (Fig. 2C), and animal models showing robust leukocyte infiltration in brain after trauma (Clark et al., 1994; Carlos et al., 1997). Dual label studies suggest that astrocytes are a prominent source of caspase-7 after TBI (Fig. 2E). Neurons also expressed caspase-7, but were less prevalent than astrocytes (Fig. 2F). There were also many cells that did not immunostain for NeuN or GFAP and had the morphologic appearance of glia, possibly being microglia or oligodendrocytes. These data are consistent with the report by Larner et al. (2005), where dual-label immunohistochemistry showed robust caspase-7 immunoreactivity in GFAP-positive cells, whereas colocalization between caspase-7 and NeuN appeared present, but far less frequent based on the data presented. Another possibility is that the RT-PCR and/or antibody-based assays for caspase-7 may be misrepresentative; however, all standard controls were utilized and these assays are well established in our laboratory.

Activated caspase-7 cleaves target proteins at Asp-Glu-Val-Asp peptide motifs similar to caspase-3 (Duan et al., 1996). However, recent studies suggest that the function of caspase-7 may be different from that of caspase-3. For example, unique caspase-7 substrates have been identified with important roles in protein synthesis and degradation. One such example is dsRNA-activated protein kinase (PKR). Cleavage of PKR by caspase-7 can result in dephosphorylation of the alpha-subunit of the eukaryotic translation initiation factor-2, resulting in inhibition of protein synthesis (Saelens et al., 2001). The proteasome activator 28 subunit, PA28gamma, was identified as a caspase-7 substrate using a yeast two-hybrid system (Araya et al., 2002). In addition, other potential caspase-7 substrates include protein kinase C epsilon (Basu et al., 2002) and endothelial monocyte-activating polypeptide II (Behrendorf et al., 2000).

Furthermore, in contrast to caspase-3, a direct role for caspase-7 in programmed cell death in neurons remains controversial. There is evidence that caspase-7 is capable of inducing apoptosis in spinal cord neurons in a transgenic model of amyotrophic lateral sclerosis (Guegan et al., 2001), and increased neuronal caspase-7 expression has been reported after TBI in rats (Larner et al., 2005). However, direct injection of active caspase-7

into cultured human primary neurons fails to induce apoptosis (Zhang et al., 2000). Caspase-7 may have an expanded role in programmed cell death in the absence of caspase-3, although this role may be species dependent, operative in mice but not humans, and primarily compensatory (Houde et al., 2004). Finally, it is unclear whether the expression of caspase-7 in astrocytes is promoting apoptotic cell death or serving a signaling function as discussed above. The importance of astrocytes in intercellular signaling has recently come to light (Fellin and Carmignoto, 2004), and death of astrocytes after TBI has also been found to be more prominent than previously recognized (Zhou et al., 2003).

In conclusion, these data suggest a role for caspase-7 after TBI in humans. This role may be related, unrelated, or indirectly related to programmed cell death in neurons after TBI in humans. While there are some intrinsic limitations related to the use of clinical materials, there is also an important need to verify processes identified in experimental models of TBI, particularly those with therapeutic implications, in human tissues. The pros and cons of this type of study have been discussed previously (Clark et al., 1999; Kochanek et al., 2000). On the other hand, these clinical data suggest that characterization of the functional consequences of caspase-7 activation in experimental models of TBI is warranted.

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